

# Can Machine Learning Variable Selection and Individual Commodity Fundamentals Improve Futures Return Forecasting?

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**DRAFT**

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## Abstract

# Contents

|          |                                                                 |           |
|----------|-----------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                             | <b>4</b>  |
| 1.1      | Motivation & Background . . . . .                               | 4         |
| 1.2      | The Research Gap . . . . .                                      | 4         |
| 1.3      | Research Questions & Hypotheses . . . . .                       | 4         |
| 1.4      | Contributions to the Literature . . . . .                       | 4         |
| 1.5      | Thesis Outline . . . . .                                        | 4         |
| <b>2</b> | <b>Literature Review</b>                                        | <b>5</b>  |
| 2.1      | Theories of Commodity Futures Returns . . . . .                 | 5         |
| 2.2      | Aggregate Commodity-Specific Factors . . . . .                  | 7         |
| 2.3      | Individual Commodity Fundamentals . . . . .                     | 7         |
| 2.4      | Macroeconomic Predictors . . . . .                              | 7         |
| 2.5      | ML Methods in Return Forecasting . . . . .                      | 7         |
| 2.6      | Volatility Regimes and Predictability . . . . .                 | 7         |
| <b>3</b> | <b>Methodology</b>                                              | <b>9</b>  |
| 3.1      | Baseline Framework (Replication) . . . . .                      | 9         |
| 3.2      | Extension I: Penalized Regression . . . . .                     | 10        |
| 3.3      | Extension II: Individual Fundamental Predictors . . . . .       | 10        |
| 3.4      | OOS Evaluation Criteria . . . . .                               | 10        |
| 3.5      | Portfolio Back-Testing Framework . . . . .                      | 10        |
| <b>4</b> | <b>Data</b>                                                     | <b>11</b> |
| 4.1      | Commodity Futures Returns . . . . .                             | 11        |
| 4.2      | Macroeconomic Factors . . . . .                                 | 11        |
| 4.3      | Aggregate Commodity-Specific Factors . . . . .                  | 11        |
| 4.4      | Individual Fundamental Predictors . . . . .                     | 11        |
| 4.5      | Sample Splits and Descriptive Statistics . . . . .              | 11        |
| <b>5</b> | <b>Replication &amp; Baseline Results</b>                       | <b>12</b> |
| 5.1      | In-Sample Estimation (1989-2003) . . . . .                      | 12        |
| 5.2      | OOS Predictive Accuracy (2004-2012) . . . . .                   | 12        |
| 5.3      | Portfolio Back-Testing (2004-2012) . . . . .                    | 12        |
| 5.4      | Replication Assessment . . . . .                                | 12        |
| <b>6</b> | <b>Extension I: ML Variable Selection</b>                       | <b>13</b> |
| 6.1      | Model Specification and Implementation . . . . .                | 13        |
| 6.2      | Variable Selection Patterns . . . . .                           | 13        |
| 6.3      | OOS Forecasting Results - Original Sample (2004-2012) . . . . . | 13        |
| 6.4      | OOS Forecasting Results - Extended Sample (2013-2024) . . . . . | 13        |
| 6.5      | Portfolio Back-Testing . . . . .                                | 13        |
| 6.6      | Robustness Checks . . . . .                                     | 13        |

|          |                                                                   |           |
|----------|-------------------------------------------------------------------|-----------|
| <b>7</b> | <b>Extension II: Individual Commodity Fundamentals</b>            | <b>14</b> |
| 7.1      | Predictor Construction by Commodity Group . . . . .               | 14        |
| 7.2      | Individual vs. Aggregate Factors: Are They Substitutes? . . . . . | 14        |
| 7.3      | OOS Forecasting Results (2004-2024) . . . . .                     | 14        |
| 7.4      | Combined Model . . . . .                                          | 14        |
| 7.5      | Portfolio Back-Testing . . . . .                                  | 14        |
| 7.6      | Regime Interaction . . . . .                                      | 14        |
| <b>8</b> | <b>Synthesis &amp; Discussion</b>                                 | <b>15</b> |
| 8.1      | Model Horse Race . . . . .                                        | 15        |
| 8.2      | Statistical vs. Economic Loss Functions Revisited . . . . .       | 15        |
| 8.3      | Implications for Commodity Market Segmentation . . . . .          | 15        |
| 8.4      | Practical Implications . . . . .                                  | 15        |
| <b>9</b> | <b>Conclusions</b>                                                | <b>16</b> |
| 9.1      | Summary of Findings . . . . .                                     | 16        |
| 9.2      | Limitations . . . . .                                             | 16        |
| 9.3      | Future Research Directions . . . . .                              | 16        |
|          | <b>References</b>                                                 | <b>17</b> |
|          | <b>Appendix</b>                                                   | <b>18</b> |

# 1 Introduction

## 1.1 Motivation & Background

(Guidolin and Pedio, 2021)

*The financialization of commodity markets; why forecasting commodity futures matters to investors, hedgers, and risk managers; the tension between statistical and economic loss functions.*

## 1.2 The Research Gap

*Three specific limitations of Guidolin & Pedio (2021): fragility of stepwise AIC selection, use of only aggregate commodity factors rather than individual fundamentals, and a sample that ends in 2012 — missing the COVID shock and post-GFC cycles.*

## 1.3 Research Questions & Hypotheses

This thesis aims at answering whether replacing AIC-based stepwise regression with penalized ML methods (LASSO/Elastic Net) improve the out-of-sample forecasting accuracy of commodity futures returns, both statistically and in portfolio returns. Furthermore, it assesses if commodity-specific individual fundamental predictors (inventory levels, crop reports, weather data) add forecasting power beyond aggregate cross sectional factors on an extended 2004-2024 sample.

## 1.4 Contributions to the Literature

*Methodological contribution (ML selection), empirical contribution (individual fundamentals), and out-of-sample robustness contribution (extended sample).*

## 1.5 Thesis Outline

Explain the structure of the thesis.

## 2 Literature Review

### 2.1 Theories of Commodity Futures Returns

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| Lag | ACF_lower | ACF      | ACF_upper |
|-----|-----------|----------|-----------|
| 1   | 0.973131  | 0.999222 | 1.025313  |
| 2   | 0.953351  | 0.998518 | 1.043686  |
| 3   | 0.939523  | 0.997812 | 1.056100  |
| 4   | 0.928182  | 0.997124 | 1.066067  |
| 5   | 0.918358  | 0.996504 | 1.074650  |
| 6   | 0.909523  | 0.995888 | 1.082252  |
| 7   | 0.901448  | 0.995305 | 1.089161  |
| 8   | 0.893894  | 0.994680 | 1.095466  |
| 9   | 0.886786  | 0.994047 | 1.101308  |
| 10  | 0.879995  | 0.993353 | 1.106712  |
| 11  | 0.873548  | 0.992686 | 1.111823  |
| 12  | 0.867391  | 0.992032 | 1.116673  |
| 13  | 0.861472  | 0.991377 | 1.121282  |
| 14  | 0.855780  | 0.990737 | 1.125694  |
| 15  | 0.850308  | 0.990128 | 1.129949  |

**Table 1:** ACF comparison (with bounds)

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|                               | PG US Equity |
|-------------------------------|--------------|
| count                         | 5642.000000  |
| mean                          | 0.000210     |
| std                           | 0.011285     |
| min                           | -0.091428    |
| 25%                           | -0.005063    |
| 50%                           | 0.000354     |
| 75%                           | 0.005859     |
| max                           | 0.113409     |
| skewness                      | -0.149761    |
| kurtosis                      | 12.483725    |
| variance                      | 0.000127     |
| median                        | 0.000354     |
| mode                          | 0.000000     |
| sum                           | 1.183665     |
| sum of squared deviations     | 0.718408     |
| jarque-bera                   | 21122.533522 |
| jarque-bera probability value | 0.000000     |

**Table 2:** Summary statistics

## 2.2 Aggregate Commodity-Specific Factors

| Model     | AIC           | BIC           | HQIC          |
|-----------|---------------|---------------|---------------|
| CER       | -34588.171062 | -34581.533068 | -34585.858721 |
| ARMA(3,2) | -34626.324616 | -34579.862382 | -34610.139093 |
| ARMA(5,5) | -34620.963843 | -34541.320684 | -34593.218710 |
| ARMA(0,5) | -34621.488221 | -34575.029712 | -34605.303561 |

**Table 3:** Model Comparison: Information Criteria

## 2.3 Individual Commodity Fundamentals

## 2.4 Macroeconomic Predictors

## 2.5 ML Methods in Return Forecasting

## 2.6 Volatility Regimes and Predictability

As you can see in Table 5.

| Series         | Test | Statistic | p-value  | Conclusion                       | Suggests         |
|----------------|------|-----------|----------|----------------------------------|------------------|
| $Y_t$          | ADF  | -0.8753   | 0.7960   | Fail to reject H0 (unit root)    | non-stationarity |
|                | PP   | -1.0501   | 0.7346   | Fail to reject H0 (unit root)    | non-stationarity |
|                | KPSS | 0.3898    | < 0.0816 | Fail to reject H0 (stationarity) | stationarity     |
| $\Delta Y_t$   | ADF  | -16.6776  | 0.0000   | Reject H0 (unit root)            | stationarity     |
|                | PP   | -16.8665  | 0.0000   | Reject H0 (unit root)            | stationarity     |
|                | KPSS | 0.2874    | < 0.1000 | Fail to reject H0 (stationarity) | stationarity     |
| $\Delta^2 Y_t$ | KPSS | 0.0207    | > 0.1000 | Fail to reject H0 (stationarity) | stationarity     |

**Table 4:** Stationarity Tests on Dividend Series

| Model       | RMSE  | R2    |
|-------------|-------|-------|
| OLS         | 0.421 | 0.120 |
| LASSO       | 0.387 | 0.180 |
| Elastic Net | 0.372 | 0.210 |

**Table 5:** Model comparison



## 3 Methodology

### 3.1 Baseline Framework (Replication)

We begin by gathering all  $T$  observations of  $n$  variables into a data matrix  $\mathbf{X} \in \mathbb{R}^{T \times n}$ , where each row is a *month* and each column is a macroeconomic variable. Each column is demeaned and standardized to unit variance using the StandardScaler:

$$X_{\text{scaled},ij} = \frac{X_{\text{raw},ij} - \mu_j}{\sigma_j}. \quad (1)$$

Let  $\Sigma = V[\mathbf{X}] \in \mathbb{R}^{n \times n}$  denote the variance-covariance matrix of  $\mathbf{X}$ , which is calculated as:

$$\Sigma = \frac{1}{T-1} \mathbf{X}'\mathbf{X}, \quad (2)$$

and let  $\Sigma = \mathbf{U}\mathbf{\Lambda}\mathbf{U}'$  be its spectral decomposition, where  $\mathbf{U} \in \mathbb{R}^{n \times n}$  is an orthogonal matrix of order  $n$ , whose columns are the eigenvectors  $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$  of  $\Sigma$ , and  $\mathbf{\Lambda} \in \mathbb{R}^{n \times n}$  is a diagonal matrix of the same order, whose diagonal entries are the corresponding eigenvalues  $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ , which, by convention, are sorted from the largest to the smallest. By multiplying the original scaled data matrix  $\mathbf{X}$  with  $\mathbf{U}$ , we derive the principal components of  $\mathbf{X}$  as the elements of the matrix  $\mathbf{F} \in \mathbb{R}^{T \times n}$ , or formally:

$$\mathbf{F} = \mathbf{X}\mathbf{U}. \quad (3)$$

To assess how much of the total variance in the original  $n$  variables is captured by the  $k$ -th principal component, or alternatively by the first  $K$  principal components, we can compute the variance-covariance matrix of  $\mathbf{F}$  as:

$$\begin{aligned} V[\mathbf{F}] &= V[\mathbf{X}\mathbf{U}] \\ &= \mathbf{U}'\Sigma\mathbf{U} \\ &= \mathbf{U}'\mathbf{U}\mathbf{\Lambda}\mathbf{U}'\mathbf{U} \\ &= \mathbf{I}_n\mathbf{\Lambda}\mathbf{I}_n \\ &= \mathbf{\Lambda}, \end{aligned} \quad (4)$$

where we exploited the orthogonality of  $\mathbf{U}$ . Recalling the fact that  $\mathbf{\Lambda}$  is a diagonal matrix,  $V[\mathbf{F}] = \mathbf{\Lambda}$  implies that the principal components are mutually uncorrelated, and have a variance equal to the associated eigenvalue. Therefore, the proportion of variance in the original  $n$  variables explained by the  $k$ -th principal component is given by:

$$\frac{\lambda_k}{\sum_{i=1}^n \lambda_i}, \quad (5)$$

and the cumulative proportion of explained variance by the first  $K$  components by:

$$\frac{\sum_{k=1}^K \lambda_k}{\sum_{i=1}^n \lambda_i}. \quad (6)$$

Mirroring the approach used by Guidolin and Pedio, 2021, we forecast the returns of each commodity  $j$ ,  $R_{t+1,j}$ , with the principal component analysis to summarize a large set of macroeconomic factors:

$$R_{t+1,j} = \underbrace{\alpha_j}_{\text{intercept}} + \underbrace{\sum_{i=1}^N \beta_{ij} PC_{t,ij}}_{\text{macroeconomic}} + \underbrace{D_{t,j} \left( \sum_{k=1}^K \gamma_{kj} C_{t,kj} \right)}_{\text{commodity-specific}} + \underbrace{\varepsilon_{t,j}}_{\text{error term}} \quad (7)$$

where  $\alpha_j$  is the model's intercept,

### 3.2 Extension I: Penalized Regression

In the first alternate approach, we keep the principal component analysis and still extract ten principal components from the  $n$  initial variables, but instead of running stepwise AIC on those selected principal components, we run a LASSO regression as an alternative way to further reduce the number of parameters to be estimated. With this only being a narrow replacement compared to the original framework, every difference in results should be attributable solely to the selection method.

$$R_{t+1,j} = \alpha_j + \sum_{i=1}^{10} \beta_{ij}^{\text{LASSO}} PC_{t,ij} + \sum_{k=1}^K \gamma_{kj}^{\text{LASSO}} C_{t,kj} + \varepsilon_{t,j} \quad (8)$$

A broader extension to the approach by Guidolin and Pedio, 2021 is foregoing the principal component analysis entirely and applying LASSO directly to the  $n$  variables.

$$R_{t+1,j} = \alpha_j + \sum_{i=1}^{128} \beta_{ij}^{\text{LASSO}} X_{t,i} + \sum_{k=1}^K \gamma_{kj}^{\text{LASSO}} C_{t,kj} + \varepsilon_{t,j} \quad (9)$$

### 3.3 Extension II: Individual Fundamental Predictors

### 3.4 OOS Evaluation Criteria

### 3.5 Portfolio Back-Testing Framework

## **4 Data**

- 4.1 Commodity Futures Returns**
- 4.2 Macroeconomic Factors**
- 4.3 Aggregate Commodity-Specific Factors**
- 4.4 Individual Fundamental Predictors**
- 4.5 Sample Splits and Descriptive Statistics**

## **5 Replication & Baseline Results**

**5.1 In-Sample Estimation (1989-2003)**

**5.2 OOS Predictive Accuracy (2004-2012)**

**5.3 Portfolio Back-Testing (2004-2012)**

**5.4 Replication Assessment**

## **6 Extension I: ML Variable Selection**

### **6.1 Model Specification and Implementation**

### **6.2 Variable Selection Patterns**

### **6.3 OOS Forecasting Results - Original Sample (2004-2012)**

### **6.4 OOS Forecasting Results - Extended Sample (2013-2024)**

### **6.5 Portfolio Back-Testing**

### **6.6 Robustness Checks**

## **7 Extension II: Individual Commodity Fundamentals**

**7.1 Predictor Construction by Commodity Group**

**7.2 Individual vs. Aggregate Factors: Are They Substitutes?**

**7.3 OOS Forecasting Results (2004-2024)**

**7.4 Combined Model**

**7.5 Portfolio Back-Testing**

**7.6 Regime Interaction**

## **8 Synthesis & Discussion**

### **8.1 Model Horse Race**

### **8.2 Statistical vs. Economic Loss Functions Revisited**

### **8.3 Implications for Commodity Market Segmentation**

### **8.4 Practical Implications**

## **9 Conclusions**

### **9.1 Summary of Findings**

### **9.2 Limitations**

### **9.3 Future Research Directions**



## References

Guidolin, Massimo and Manuela Pedio (Apr. 2021). “Forecasting Commodity Futures Returns with Stepwise Regressions: Do Commodity-Specific Factors Help?” In: *Annals of Operations Research* 299.1–2, pp. 1317–1356. ISSN: 0254-5330, 1572-9338. DOI: 10.1007/s10479-020-03515-w. URL: <http://link.springer.com/10.1007/s10479-020-03515-w> (visited on 04/17/2026).

## Appendix